

MATH211

Discrete Mathematics

Matrices & Integers



Agenda

2.6 Matrices

4.3 Primes and Greatest Common Divisors

Discrete Mathematics and Its Applications

Kenneth H. Rosen

2.6. Matrices



- Matrix arithmetic
- Matrix multiplication
- Transpose
- Powers of matrices
- Zero one matrices

Applications of Matrices

Tons of applications, including:

- ▶ Solving systems of linear equations
- ▶ Computer Graphics, Image Processing
- ▶ Models within Computational Science & Engineering
- ▶ Quantum Mechanics, Quantum Computing
- ▶ Many, many more...

Matrices

- ▶ A *matrix* is a rectangular array of objects (usually numbers).
- ▶ An $m \times n$ (“ m by n ”) matrix has exactly m horizontal rows and n vertical columns.
- ▶ An $n \times n$ matrix is called a **square** matrix, whose *order* is n .

Note: The plural of “matrix” is “*matrices*”

Row $\longrightarrow$

$$\begin{bmatrix} 2 & 3 \\ 5 & -1 \\ 7 & 0 \end{bmatrix}$$

Column $\uparrow$

3×2 matrix

Row and Column Order

- ▶ The **rows** in a matrix are usually indexed **1** to **m** from **top to bottom**.
- ▶ The **columns** are usually indexed **1** to **n** from **left to right**.
- ▶ Elements are indexed by row, then column.

$$\mathbf{A} = [a_{i,j}] = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

↗ ↖
Row Column

Matrix Equality

- ▶ Two matrices **A** and **B** are equal if and only if they have the same number of rows, the same number of columns, and all corresponding elements are equal.

$$\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix} \neq \begin{bmatrix} 3 & 2 & 0 \\ -1 & 6 & 0 \end{bmatrix}$$

Matrix Sums

- ▶ The *sum* $\mathbf{A+B}$ of two matrices **A** and **B** (which **must** have the same number of rows, and the same number of columns) is the matrix (also with the same shape) given by adding corresponding elements.
- ▶ $\mathbf{A+B} = [a_{i,j}+b_{i,j}]$

$$\begin{bmatrix} 2 & 6 \\ 0 & -8 \end{bmatrix} + \begin{bmatrix} 9 & 3 \\ -11 & 3 \end{bmatrix} =$$

Matrix Sums

- ▶ The *sum* $\mathbf{A+B}$ of two matrices **A** and **B** (which **must** have the same number of rows, and the same number of columns) is the matrix (also with the same shape) given by adding corresponding elements.
- ▶ $\mathbf{A+B} = [a_{i,j}+b_{i,j}]$

$$\begin{bmatrix} 2 & 6 \\ 0 & -8 \end{bmatrix} + \begin{bmatrix} 9 & 3 \\ -11 & 3 \end{bmatrix} = \begin{bmatrix} 11 & 9 \\ -11 & -5 \end{bmatrix}$$

Matrix Products

- For an $m \times k$ matrix $\mathbf{A}$ and a $k \times n$ matrix $\mathbf{B}$, the *product* $\mathbf{AB}$ is the $m \times n$ matrix:

$$\mathbf{AB} = \mathbf{C} = [c_{i,j}] \equiv \left[\sum_{\ell=1}^k a_{i,\ell} b_{\ell,j} \right]$$

- i.e., element (i,j) of $\mathbf{AB}$ is given by the *vector dot product* of the i th row of $\mathbf{A}$ and the j th column of $\mathbf{B}$ (considered as vectors).

"Dot Product"

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & \end{bmatrix}$$

Matrix Product Example 1

$$\begin{pmatrix} \overrightarrow{4} & \overrightarrow{2} & \overrightarrow{4} \\ \overrightarrow{8} & \overrightarrow{3} & \overrightarrow{1} \end{pmatrix} \cdot \begin{pmatrix} \downarrow 3 & \downarrow 5 \\ \downarrow 2 & \downarrow 8 \\ \downarrow 7 & \downarrow 9 \end{pmatrix} = \begin{pmatrix} \color{red}{44} & \color{black}{72} \\ \color{teal}{37} & \color{violet}{73} \end{pmatrix}$$

$$\color{red}{\rightarrow} \quad 4 \times 3 + 2 \times 2 + 4 \times 7 = \color{red}{44}$$

$$\color{black}{\rightarrow} \quad 4 \times 5 + 2 \times 8 + 4 \times 9 = \color{black}{72}$$

$$\color{teal}{\rightarrow} \quad 8 \times 3 + 3 \times 2 + 1 \times 7 = \color{teal}{37}$$

$$\color{violet}{\rightarrow} \quad 8 \times 5 + 3 \times 8 + 1 \times 9 = \color{violet}{73}$$

Matrix Product Example 2

- An example matrix multiplication to practice in class:

$$\begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 & 1 & 0 \\ 2 & 0 & -2 & 0 \\ 1 & 0 & 3 & 1 \end{bmatrix}$$

Matrix Product Example 2

- An example matrix multiplication to practice in class:

$$\begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 & -1 & 1 & 0 \\ 2 & 0 & -2 & 0 \\ 1 & 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -5 & -1 \\ 3 & -2 & 11 & 3 \end{bmatrix}$$

Matrix Multiplication is **not** Commutative

► Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

Does $\mathbf{AB} = \mathbf{BA}$?

Matrix Multiplication is **not** Commutative

► Let $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

Does $\mathbf{AB} = \mathbf{BA}$?

$$\mathbf{AB} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{BA} = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

Hence, $\mathbf{AB} \neq \mathbf{BA}$.

Identity Matrices

- ▶ $A.I = A$
- ▶ The identity matrix is a **square** matrix ($n \times n$)
- ▶ $I_n = \delta_{ij}$ where $\delta_{ij}=1$ if $i = j$ and $\delta_{ij}=0$ if $i \neq j$ (diagonal is 1's and all other elements are 0).

$$\mathbf{I}_n = \begin{bmatrix} \mathbf{1} \text{ if } i = j \\ \mathbf{0} \text{ if } i \neq j \end{bmatrix} = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{1} \end{bmatrix}$$

Example

► Find $A.I$ where $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -2 & 7 \\ 0 & 1 & -1 \end{bmatrix}$

$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & -2 & 7 \\ 0 & 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -2 & 7 \\ 0 & 1 & -1 \end{bmatrix}$$

Powers of Matrices

If $\mathbf{A}$ is an $n \times n$ square matrix and $p \geq 0$, then:

▶ $\mathbf{A}^p \equiv \underbrace{\mathbf{A} \mathbf{A} \mathbf{A} \cdots \mathbf{A}}_{p \text{ times}}$

▶ $\mathbf{A}^2 = \mathbf{A} \cdot \mathbf{A}$

▶ $\mathbf{A}^3 = \mathbf{A}^2 \cdot \mathbf{A}$

▶ Example: $\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}^3$

Powers of Matrices

$$\begin{aligned}\begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}^3 &= \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 2 \\ -2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix}\end{aligned}$$

Transpose of Matrices

- ▶ The transpose of $A_{m \times n}$ denoted A^t is the $n \times m$ matrix obtained by *interchanging the rows and columns* of A.

- ▶ $A = \begin{bmatrix} 4 & 5 & 7 \\ 0 & 1 & -1 \end{bmatrix}$ $A^t = \begin{bmatrix} 4 & 0 \\ 5 & 1 \\ 7 & -1 \end{bmatrix}$

- ▶ **If $A = A^t$ then A is symmetric**

- ▶ $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $A^t = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

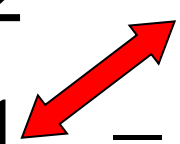
Symmetric

Symmetric Matrices

- ▶ A square matrix $\mathbf{A}$ is *symmetric* iff $\mathbf{A} = \mathbf{A}^T$ i.e., $\forall i, j \leq n: a_{ij} = a_{ji}$
- ▶ Which is symmetric?

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 1 & 3 \\ 1 & 0 & -1 \\ 3 & -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & -2 \end{bmatrix}$$


Zero-One Matrices

- ▶ All elements of a zero-one matrix are 0 or 1.
- ▶ It is based on Boolean operations.
 - ▶ Representing **False** & **True** respectively
- ▶ The **meet** of **A**, **B** (both $m \times n$ zero-one matrices):
 - ▶ $\mathbf{A} \wedge \mathbf{B} \equiv [a_{ij} \wedge b_{ij}] = [a_{ij} b_{ij}]$
- ▶ The **join** of **A**, **B**:
 - ▶ $\mathbf{A} \vee \mathbf{B} \equiv [a_{ij} \vee b_{ij}]$

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise,} \end{cases}$$

$$b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Example

- ▶ Find the join and meet of the zero-one matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

- ▶ The join of A and B is:

$$\mathbf{A} \vee \mathbf{B} = \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

- ▶ The meet of A and B is:

$$\mathbf{A} \wedge \mathbf{B} = \begin{bmatrix} 1 \wedge 0 & 0 \wedge 1 & 1 \wedge 0 \\ 0 \wedge 1 & 1 \wedge 1 & 0 \wedge 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Boolean Products

- ▶ Let $\mathbf{A}=[a_{ij}]$ be an $m \times k$ zero-one matrix, & let $\mathbf{B}=[b_{ij}]$ be a $k \times n$ zero-one matrix,
- ▶ The *boolean product* of $\mathbf{A}$ and $\mathbf{B}$ is like normal matrix $\times$, but using $\vee$ instead $+$ in the row-column “vector dot product.”

$$\mathbf{AB} = \mathbf{C} = [c_{i,j}] \equiv \left[\sum_{\ell=1}^k a_{i,\ell} b_{\ell,j} \right]$$

$$\mathbf{A} \odot \mathbf{B} = \mathbf{C} = [c_{ij}] = \left[\bigvee_{\ell=1}^k a_{i\ell} \wedge b_{\ell j} \right]$$

Example

- Find the Boolean product of A and B where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Solution:

$$\begin{aligned} \mathbf{A} \odot \mathbf{B} &= \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix} \\ &= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \end{aligned}$$

Boolean Powers

- ▶ For a square zero-one matrix $\mathbf{A}$, and any $k \geq 0$, the k^{th} Boolean power of $\mathbf{A}$ is simply the Boolean product of k copies of $\mathbf{A}$.

- ▶ $\mathbf{A}^{[k]} \equiv \underbrace{\mathbf{A} \odot \mathbf{A} \odot \dots \odot \mathbf{A}}_{k \text{ times}}$

Example

► Let $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ Find $A^{[n]}$ for all positive integers n .

Solution:

Example

► Let $\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ Find $\mathbf{A}^{[n]}$ for all positive integers n .

Solution:

$$\mathbf{A}^{[2]} = \mathbf{A} \odot \mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\mathbf{A}^{[3]} = \mathbf{A}^{[2]} \odot \mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{A}^{[4]} = \mathbf{A}^{[3]} \odot \mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\mathbf{A}^{[5]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

We can now see that $\mathbf{A}^{[n]} = \mathbf{A}^{[5]}$ for all positive integers n with $n \geq 5$.

4.3 Primes and Greatest Common Divisors



- Prime Numbers
- Greatest Common Divisors
- The Euclidian Algorithm

Integer Division

- ▶ If a and b are integers with $b \neq 0$, we say that b divides a (a is divided by b) if there is an integer k such that $a = bk$.
- ▶ When b divides a we say that b is a factor of a and that a is a multiple of b .

$$\frac{a}{b}$$

- ▶ The notation $b \mid a$ (a / b) denotes that b divides a .

$$\frac{a}{b} = \frac{8}{2}$$

- ▶ We write $b \nmid a$ when b does not divide a .

$$\frac{a}{b} = \frac{9}{2}$$

Examples

▶ $4 \mid 12$ $4 \nmid 18$

▶ $3 \nmid 7$, since $7/3$ is not an integer.

Primes

- ▶ A positive integer p greater than 1 is called **prime** if the only positive factors of p are **1** and p . A prime number is a number that can only be divided by itself and 1 without remainders
- ▶ A positive integer that is greater than 1 and is **not prime** is called **composite**.

Example:

- ▶ 7 is prime because its only positive factors are 1 and 7
- ▶ 9 is composite because it is divisible by 3

The Fundamental Theorem of Arithmetic [Inspection: By Converting the Factors into Prime Numbers Only]

Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of non-decreasing size.

Examples:

- $100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$
- $641 = 641$
- $999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$
- $1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$

$$\begin{aligned} 20 &= 4 \times 5 \\ &= 2 \times 2 \times 5 = 2^2 \cdot 5 \end{aligned}$$

$$\begin{aligned} 100 &= 4 \times 25 \\ &= 2 \times 2 \times 5 \times 5 = 2^2 \cdot 5^2 \end{aligned}$$

The Fundamental Theorem of Arithmetic

[By Division by Prime Numbers]

- Find the prime factorization of the given numbers: 216 and 234

$$\begin{array}{r|l}
 216 & 2 \\
 108 & 2 \\
 54 & 2 \\
 27 & 3 \\
 9 & 3 \\
 3 & 3 \\
 1 &
 \end{array}$$

$$\begin{array}{r|l}
 234 & 2 \\
 117 & 3 \\
 39 & 3 \\
 13 & 13 \\
 1 &
 \end{array}$$

$$216 = 2^3 \cdot 3^3$$

$$234 = 2 \cdot 3^2 \cdot 13$$

Example

- Find the prime factorization of the given numbers: 135 and 351

$$\begin{array}{r|l}
 135 & 5 \\
 27 & 3 \\
 9 & 3 \\
 3 & 3 \\
 1 &
 \end{array}$$

$$\begin{array}{r|l}
 351 & 3 \\
 117 & 3 \\
 39 & 3 \\
 13 & 13 \\
 1 &
 \end{array}$$

$$135 = 5 \cdot 3^3$$

$$351 = 3^3 \cdot 13$$

Finding the Greatest Common Divisor By Inspection

Definition: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and also $d \mid b$ is called the greatest common divisor of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$. It is the largest integer d that both numbers (a and b) can be divided by it.

► What is the greatest common divisor of 24 and 36?

► **Solution:** $\gcd(24, 36) = 12$

$$\frac{24}{36}$$

► What is the greatest common divisor of 17 and 22?

► **Solution:** $\gcd(17, 22) = 1$

$$\frac{17}{22}$$

Finding the Greatest Common Divisor Using Prime Factorizations

37

1. Find the prime factorization of each number
2. Select all the primes that are common to all the factorizations and apply to each one the smallest exponent to which it occurs
3. The product of the factors selected in step 2 is the GCD of all the numbers.

Note: It can be called HCF (Highest Common Factor)

Example

- Find the GCD of the given numbers: 216 and 234

$$\begin{array}{r|l}
 216 & 2 \\
 108 & 2 \\
 54 & 2 \\
 27 & 3 \\
 9 & 3 \\
 3 & 3 \\
 1 &
 \end{array}$$

$$\begin{array}{r|l}
 234 & 2 \\
 117 & 3 \\
 39 & 3 \\
 13 & 13 \\
 1 &
 \end{array}$$

$$216 = 2^3 \cdot 3^3$$

$$234 = 2 \cdot 3^2 \cdot 13$$

$$\text{GCD}(216, 234) = 2 \cdot 3^2 = 18$$

Example

- Find the GCD of the given numbers: 135 and 351

$$\begin{array}{r|l}
 135 & 5 \\
 27 & 3 \\
 9 & 3 \\
 3 & 3 \\
 1 &
 \end{array}
 \qquad
 \begin{array}{r|l}
 351 & 3 \\
 117 & 3 \\
 39 & 3 \\
 13 & 13 \\
 1 &
 \end{array}$$

$$135 = 5 \cdot 3^3$$

$$351 = 3^3 \cdot 13$$

$$\text{GCD}(135, 351) = 3^3 = 27$$

Example

- ▶ Find the $\text{GCD}(16, 32)$
 - ▶ $16 = 2^4$
 - ▶ $32 = 2^5 = 2^4 \cdot 2$
 - ▶ So $\text{GCD}(16, 32) = 2^4 = 16$

Example

- ▶ Find the $\text{GCD}(12, 30)$
 - ▶ $12 = 2^2 \cdot 3$
 - ▶ $30 = 2 \cdot 3 \cdot 5$
 - ▶ $\text{GCD} = 2^1 \cdot 3^1 = 6$

Relatively Prime

Definition: The integers a and b are **relatively prime** if their greatest common divisor is **1**.

Relatively prime: they don't have any factors in common.

Example: 17 and 22

Definition: The integers $a_1, a_2, \dots, a_n$ are pairwise relatively prime if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Examples

- ▶ Determine whether the integers 10, 17 and 21 are pairwise relatively prime.

Solution

- ▶ Because $\gcd(10,17) = 1$, $\gcd(10,21) = 1$, and $\gcd(17,21) = 1$.
 - ▶ Therefore, 10, 17, and 21 are pairwise relatively prime.
-
- ▶ Determine whether the integers 10, 19, and 24 are pairwise relatively prime.

Solution

- ▶ Because $\gcd(10,24) = 2$.
- ▶ Therefore, 10, 19, and 24 are not pairwise relatively prime.

Finding the Greatest Common Divisor Using Euclidean Algorithm

44

- The Euclidian algorithm is an efficient method for computing the greatest common divisor of two integers. It is based on the idea that $\gcd(a,b)$ is equal to $\gcd(b,c)$ when $a > b$ and c is the remainder when a is divided by b .

Example: Find $\gcd(91, 287)$:

If $a = b(k) + c$ then
 $\gcd(a, b) = \gcd(b, c)$

$$13 = 4(3) + 1$$

$$\gcd(13, 4) = \gcd(4, 1)$$

Finding the Greatest Common Divisor Using Euclidean Algorithm

45

- The Euclidean algorithm is an efficient method for computing the greatest common divisor of two integers. It is based on the idea that $\gcd(a, b)$ is equal to $\gcd(b, c)$ when $a > b$, and c is the remainder when a is divided by b .

Example: Find $\gcd(287, 91)$:

- $287 = 91 (3) + 14$ Divide 287 by 91
- $91 = 14 (6) + 7$ Divide 91 by 14
- $14 = 7 (2) + 0$ Divide 14 by 7

Stopping
condition

If $a = b (k) + c$ then
 $\gcd(a, b) = \gcd(b, c)$

$$\gcd(287, 91) = \gcd(91, 14) = \gcd(14, 7) = 7$$

Example

Find the $\gcd(414, 662)$

- ▶ $662 = 414(1) + 248$
- ▶ $414 = 248(1) + 166$
- ▶ $248 = 166(1) + 82$
- ▶ $166 = 82(2) + 2$
- ▶ $82 = 2(41) + 0$

$$\gcd(662, 414) = \gcd(414, 248) = \gcd(248, 166) = \gcd(166, 82) = \gcd(82, 2) = 2$$

Review Questions



Problem 1

1. Let $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -7 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 8 \\ 7 & 6 \end{bmatrix}$

Find

a) $A - B$

b) AB

c) A^2

d) BA

e) B^2

f) $2B$

Solution of Problem 1

- a) $A - B$ Not Possible
- b) AB Not Possible
- c) A^2 Not Possible

Solution of Problem 1

$$d) \quad BA = \begin{bmatrix} -2 & 8 \\ 7 & 6 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & -1 \\ 4 & -7 & 3 \end{bmatrix} = \begin{bmatrix} 0 + 32 & -2 - 56 & 1 + 24 \\ 0 + 24 & 7 - 42 & -7 + 18 \end{bmatrix} = \begin{bmatrix} 32 & -58 & 25 \\ 24 & -35 & 11 \end{bmatrix}$$

$$e) \quad B^2 = \begin{bmatrix} -2 & 8 \\ 7 & 6 \end{bmatrix} \times \begin{bmatrix} -2 & 8 \\ 7 & 6 \end{bmatrix} = \begin{bmatrix} 4 + 56 & -16 + 48 \\ -14 + 42 & 56 + 36 \end{bmatrix} = \begin{bmatrix} 60 & 32 \\ 28 & 92 \end{bmatrix}$$

$$f) \quad 2B = 2 \begin{bmatrix} -2 & 8 \\ 7 & 6 \end{bmatrix} = \begin{bmatrix} -4 & 16 \\ 14 & 12 \end{bmatrix}$$

Problem 2

- Find the GCD $(14, 0)$

The largest number that both numbers $(14, 0)$ can be divided by is 14

Problem 3

- Summation of Matrix A and Matrix B

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & -3 \\ 3 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 & -1 \\ 1 & -3 & 0 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & -2 \\ 3 & -1 & -3 \\ 2 & 5 & 2 \end{bmatrix}$$

Problem 4

► Find AB

$$A = \begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 3 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 14 & 4 \\ 8 & 9 \\ 7 & 13 \\ 8 & 2 \end{bmatrix}$$

Problem 5

- What are the greatest common divisors of these pairs of integers?

$$2^2 \cdot 3^3 \cdot 5^5 \text{ and } 2^5 \cdot 3^3 \cdot 5^2$$

$$\mathbf{GCD = 2^2 \cdot 3^3 \cdot 5^2}$$

$$2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \text{ and } 2^{11} \cdot 3^9 \cdot 11 \cdot 17^{14}$$

$$\mathbf{GCD = 2 \cdot 3 \cdot 11}$$

Problem 6

Prove that the $\gcd(252, 198) = 18$ using Euclidean Algorithm.

References

- ▶ Discrete Mathematics and Its Applications, Kenneth H. Rosen
- ▶ Michael P. Frank Slides

Supplementary Material

- ▶ Discrete Math 4.3.1 Prime Numbers and Their Properties
<https://www.youtube.com/watch?v=tJeP8882zZg>
- ▶ Discrete Math - 4.3.2 Greatest Common Divisors
<https://www.youtube.com/watch?v=zEdovJYAGwU>
- ▶ Discrete Math - 4.3.3 The Euclidean Algorithm
<https://www.youtube.com/watch?v=5LhWlaWlprM>